

NAME

instruments -- Instrument specific data files

DESCRIPTION

The **ccdred** package has been designed to accommodate many different instruments, detectors, and observatories. This is done by having instrument specific data files. Note that by instrument we mean a combination of detector, instrument, application, and observatory, so there might be several "instruments" associated with a particular CCD detector. Creating and maintaining the instrument files is generally the responsibility of the support staff, though the user may create or copy and modify his/her own instrument/application specific files. The task **setinstrument** makes this information available to the user and package easily.

There are three instrument data files, all of which are optional. The package may be used without the instrument files but much of the convenience of the package, particularly with respect to using the CCD image types, will be lost. The three files are an instrument image header translation file, an initialization task which mainly sets default task parameters, and a bad pixel file identifying the cosmic bad pixels in the detector. These files are generally stored in a system data directory which is a subdirectory of the logical directory "ccddb\$". Each file has a root name which identifies the instrument.

1. Instrument Translation File

The instrument translation file translates the parameter names used by the **ccdred** package into instrument specific parameters and also supplies instrument specific default values. The package parameter *ccdred.instrument* specifies this file to the package. The task **setinstrument** sets this parameter, though it can be set explicitly like any other parameter. For the standard instrument translation file the root name is the instrument identification and the extension is "dat" ("*.dat" files are protected from being removed in a "stripped" system, i.e. when all nonessential files are removed). Private instrument files may be given any name desired.

The instrument translation proceeds as follows. When a parameter for an image, for example "imagetyp", it looks in the instrument translation file. If the file is not found or none is specified then the image header keyword that is requested has the same name. If an instrument translation file is defined then the requested parameter is translated to an image header keyword, provided an entry is given. If no translation is given the package name is used. For example the package parameter "imagetyp" might be translated to "data-typ" (the old NOAO CCD keyword). If the parameter is not found then the default value specified in the translation file, if present, is returned. For recording parameter information in the header, such as processing flags, the translation is also used. The default value has no meaning in this case. For example, if the flag specifying that the image has been corrected by a flat field is to be set then the package parameter name "flatcor" might be translated to "ff-flag". If no translation is given then the new image header parameter is entered as "flatcor".

The format of the translation file are lines consisting of the package parameter name, followed by the image header keyword, followed by the default value. The first two fields are parameter names. The fields are separated by whitespace (blanks and tabs). String default values containing blanks must be quoted. An example is given below.

```
# Sample translation file.
exptime      itime
darktime     itime
imagetyp     data-typ
subset       flpos
biassec      biassec      [411:431,2:573]
datasec      datasec      [14:385,2:573]

fixpix       bp-flag      0
```

```

overscan    bt-flag    0
zerocor     bi-flag    0
darkcor     dk-flag    0
flatcor     ff-flag    0
fringcor    fr-flag    0

```

The first comment line is ignored as are blank lines. The first two lines translate the CCD image type, and the subset parameter without default values (see **ccdtypes** and **subsets** for more information). The next two lines give the overscan bias strip section and the data section with default values for the instrument. Note that these parameters may be overridden in the task **ccdproc**.

The next set of translations requires further discussion. For processing flags the package assumes that the absence of a keyword means that the processing has not been done. If processing is always to be done with the **CCDRED** package and no processing keywords are recorded in the raw data then these parameters should be absent (unless you don't like the names used by the package). However, for compatibility with the original NOAO CCD images, which may be processed outside of IRAF and which use 0 no processing value, the processing flags are translated and the false values are indicated by the default values.

If there is more than one translation for the same CCDRED parameter, for example more than one `exptime`, then the last one is used.

In addition to the parameter name translations the translation file contains translations between the value of the image type parameter and the image types used by the package. These lines consist of the image header type string as the first field (with quotes if there are blanks) and the image type as recognized by the package. The following example will make this clearer.

```

'OBJECT (0)'      object
'DARK (1)'        dark
'PROJECTOR FLAT (2)' flat
'SKY FLAT (3)'    other
'COMPARISON LAMP (4)' other
'BIAS (5)'        zero
'DOME FLAT (6)'   flat

```

The values of the image type strings in the header contain blanks so they are quoted. Also the case of the strings is important. Note that there are two types of flat field images and three types of object images.

The CCD image types recognized by the package are:

```

zero   - zero level image such as a bias or preflash
dark   - dark count image
flat   - flat field image
illum  - illumination image such as a sky image
fringe - fringe correction image
object - object image

```

There may be more than one image type that maps to the same package type. In particular other standard CCD image types, such as comparison spectra, multiple exposure, standard star, etc., should be mapped to object or other. There may also be more than one type of flat field, i.e. dome flat, sky flat, and lamp flat. For more on the CCD image types see **ccdtypes**.

The complete set of package parameters are given below. The package parameter names are generally the same as the standard image header keywords being adopted by NOAO.

General Image Header and Default Parameters			
ccdmean	darktime	exptime	fixfile
imagetyp	ncombine	biassec	subset
title	datasec	nscanrow	

CCDRED Processing Flags

ccdproc	darkcor	fixpix	flatcor
fringcor	illumcor	overscan	trim
zerocor			

	CCDRED CCD Image Types			
dark	flat	fringe	illum	
none	object	unknown	zero	

The translation mechanism described here may become more sophisticated in the future and a general IRAF system facility may be implemented eventually. For the present the translation mechanism is quite simple.

2. Instrument Setup Script

The task **setinstrument** translates an instrument ID into a CL script in the instrument directory. This script is then executed. Generally this script simply sets the task parameters for an instrument/application. However, it could do anything else the support staff desires. Below are the first few lines of a script.

```
ccdred.instrument = "ccddb$kpno/example.dat"
ccdred.pixeltype = "real"
ccdproc.fixpix = yes
ccdproc.overscan = yes
ccdproc.trim = yes
ccdproc.zerocor = no
ccdproc.darkcor = no
ccdproc.flatcor = yes
ccdproc.biassec = "[411:431,2:573]"
ccdproc.datasec = "[14:385,2:573]"
```

The instrument parameter should always be set unless there is no translation file for the instrument. The **ccdproc** parameters illustrate setting the appropriate processing flags for the instrument. The overscan bias and trim data sections show an alternate method of setting these instrument specific parameters. They may be set in the setup script in which case they are given explicitly in the user parameter list for **ccdproc**. If the value is "image" then the parameters may be determined either through the default value in the instrument translation file, as illustrated in the previous section, or from the image header itself.

The instrument setup script for setting default task parameters may be easily created by the support person as follows. Set the package parameters using **eparam** or with CL statements. Setting the parameters might involve testing. When satisfied with the way the package is set then the parameters may be dumped to a using the task **dparam**. The final step is editing this script to delete unimportant and query parameters. For example,

```
cl> dparam ccdred >> file.cl
cl> dparam ccdproc >> file.cl
cl> dparam combine >> file.cl
...
cl> ed file.cl
```

3. Instrument Bad Pixel File

The bad pixel file describes the bad pixels, columns, and lines in the detector which are to be replaced by interpolation when processing the images. This file is clearly detector specific. The file consists of lines describing rectangular regions of the image. The regions are specified by four numbers giving the starting and ending columns followed by the starting and ending lines. The starting and ending points may be the same to specify a single column or line. The example below illustrates a bad pixel file.

```
# RCA1 CCD untrimmed
25 25 1 512
108 108 1 512
302 302 403 512
```

```
1 512 70 70
245 246 312 315
```

If there is a comment line in the file containing the word "untrimmed" then the coordinates of the bad pixel regions apply to the original CCD detector coordinates. If the image has been trimmed and the bad pixels are replaced at a later stage then this word indicates that the trim region be determined from the image header and the necessary coordinate conversion made to the original CCD pixel coordinates. Note that if a subraster readout is used the coordinates must still refer to the original CCD coordinates and not the raw, untrimmed readout image. If the word "untrimmed" does not appear then the coordinates are assumed to apply to the image directly; i.e. the trimmed coordinates if the image has been trimmed or the original coordinates if the image has not been trimmed. The standard bad pixel files should always refer to the original, untrimmed coordinates.

The first two bad pixel regions are complete bad columns (the image is 512 x 512), the next line is a partial bad column, the next line is a bad line, and the last line is a small bad region. These files are easy to create, provided you have a good image to work from and a way to measure the positions with an image or graphics display.

SEE ALSO

ccdtypes, subsets, setinstrument